Technical Report: Periodic-1D Rank-One Single-Band Fourier Locality

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Pre-execution design

1 Scope and evidence boundary

This calculation is the proposed scalar endpoint of the controlled periodic rank ladder. It is intended to share momentum, Fourier, mesh, support, normalization, error, and evidence conventions with the rank-two and higher-rank calculations.

This report is a pre-execution design note. No scalar model coefficients, calculation input, executable stage, numerical result, convergence finding, retained run, or evidence package exists. The document does not authorize execution or evidence promotion.

2 Rank-one structural boundary

Let $H(k) \in \mathbb{C}$ be a scalar represented Hamiltonian and let $U(k) \in U(1)$. Scalar multiplication commutes, so

$$U(k)^\dagger H(k) U(k) = H(k)\, U(k)^\dagger U(k) = H(k). \tag{1}$$

Consequently, an internal phase cannot redistribute a scalar represented Hamiltonian among Fourier hopping blocks. The rank-one calculation can test scalar interpolation and truncation, but it cannot exhibit the gauge-induced represented-operator locality that becomes possible at rank two and above.

This algebraic observation determines the null-control role. It does not select the future numerical model and is not itself a finite-mesh result.

3 Required model and numerical contract

Before implementation, scientific review must select and freeze:

- an explicit scalar specialization of the controlled parent family;
- the periodic momentum domain and endpoint convention;
- a positive energy reference;
- construction, independent evaluation, and refinement meshes;
- Fourier normalization and displacement conventions;
- ordered hopping supports;
- interpolation, deliberate-truncation, and total scalar-error definitions;

- exterior-isolation qualification if an ambient parent is retained;
- binary64 tolerances and fail-closed rules; and
- strict input, result, verification, analysis, and provenance records.

No coefficients, meshes, supports, or tolerances are selected by this report.

4 Alignment-channel applicability

Only the no-alignment scalar channel is scientifically distinct. One-global-phase, pointwise-phase, known-inverse-phase, and restricted internal-rotation channels are operator-identical in rank one. They must be recorded as inapplicable as distinct alignment channels rather than emitted as duplicated numerical results.

5 Planned simulation and analysis boundary

A future governed workflow will separate:

1. experiment calculation from a frozen input;
2. producer-independent experiment verification;
3. analysis consuming only verified experiment output;
4. producer-independent analysis verification; and
5. plotting from verified analysis.

No placeholder stage is added before its input/output contract exists.

Potential analysis quantities include scalar Fourier coefficients, interpolation residuals, omitted hopping-tail weight, support-truncation errors, total evaluation-mesh errors, refinement comparisons, and first-passing support decisions under prospectively fixed tolerances.

6 Relationship to Mathieu1D

The retained `calculations/mathieu1d/` package remains a separate periodic-potential and single-band Fourier-reduction benchmark. It is neither renamed nor relabeled as the compatible rank-one specialization of this controlled family.

7 Limitations and next gate

A future passing calculation could establish only the declared finite-mesh scalar results. It would not establish higher-rank gauge behavior, continuum convergence, material applicability, or validation of Mathieu1D. The next gate is review of the explicit scalar model and its complete input/output contract.